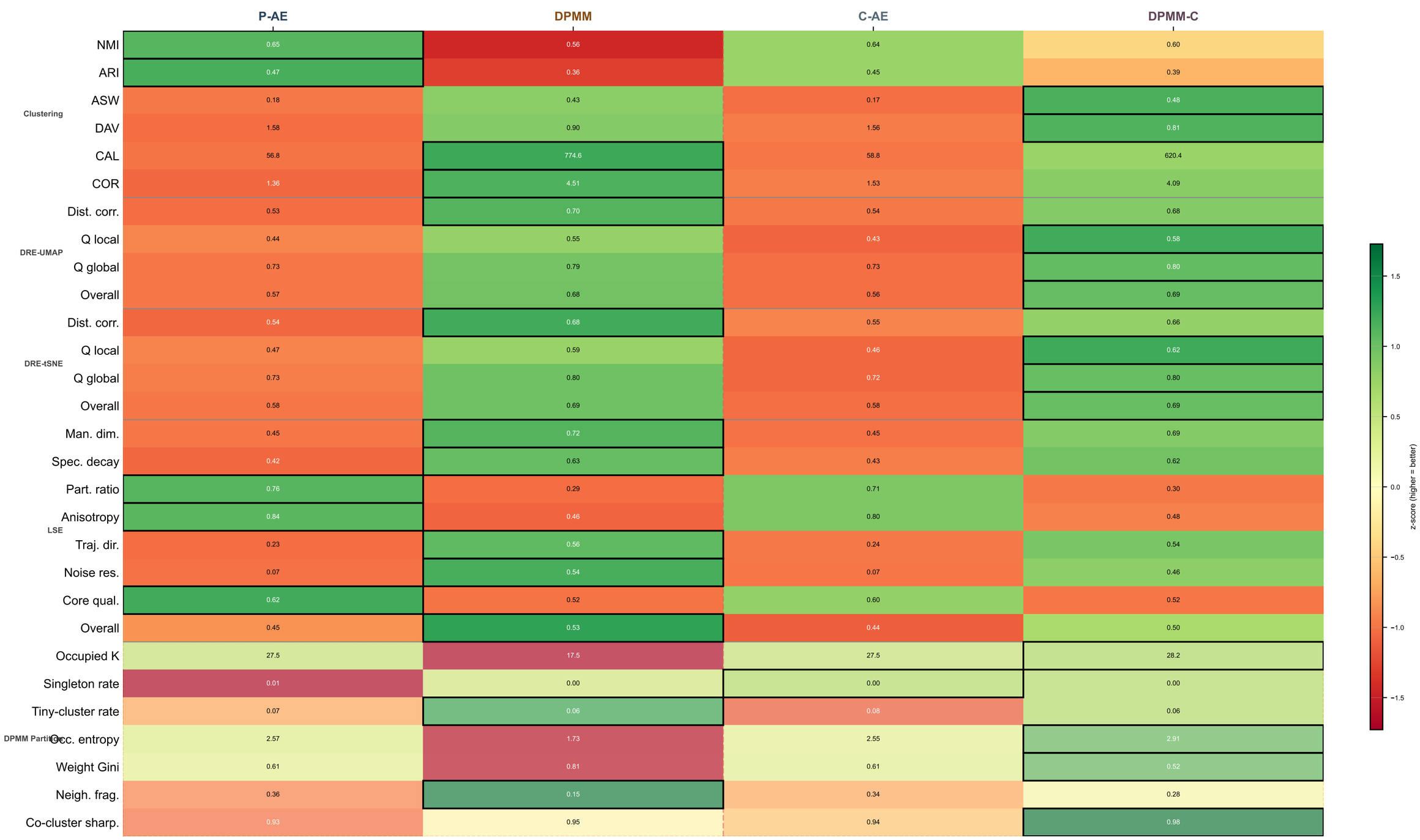
(a)

Base
Contrastive
P-AE
DPMM
C-AE
DPMM-C
setty
dentate
lung
endo

(b)


P-AE
DPMM
C-AE
DPMM-C
Clustering
NMI 0.65 0.56 0.64 0.60
ARI 0.47 0.36 0.45 0.39
ASW 0.18 0.43 0.17 0.48
DAV 1.58 0.90 1.56 0.81
CAL 56.8 774.6 58.8 620.4
COR 1.36 4.51 1.53 4.09
DRE-UMAP
Dist. corr. 0.53 0.70 0.54 0.68
Q local 0.44 0.55 0.43 0.58
Q global 0.73 0.79 0.73 0.80
Overall 0.57 0.68 0.56 0.69
DRE-tSNE
Dist. corr. 0.54 0.68 0.55 0.66
Q local 0.47 0.59 0.46 0.62
Q global 0.73 0.80 0.72 0.80
Overall 0.58 0.69 0.58 0.69
LSE
Man. dim. 0.45 0.72 0.45 0.69
Spec. decay 0.42 0.63 0.43 0.62
Part. ratio 0.76 0.29 0.71 0.30
Anisotropy 0.84 0.46 0.80 0.48
Traj. dir. 0.23 0.56 0.24 0.54
Noise res. 0.07 0.54 0.07 0.46
Core qual. 0.62 0.52 0.60 0.52
Overall 0.45 0.53 0.44 0.50
DPMM Partition
Occupied K 27.5 17.5 27.5 28.2
Singleton rate 0.01 0.00 0.00 0.00
Tiny-cluster rate 0.07 0.06 0.08 0.06
Occ. entropy 2.57 1.73 2.55 2.91
Weight Gini 0.61 0.81 0.61 0.52
Neigh. frag. 0.36 0.15 0.34 0.28
Co-cluster sharp. 0.93 0.95 0.94 0.98
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
z-score (higher = better)